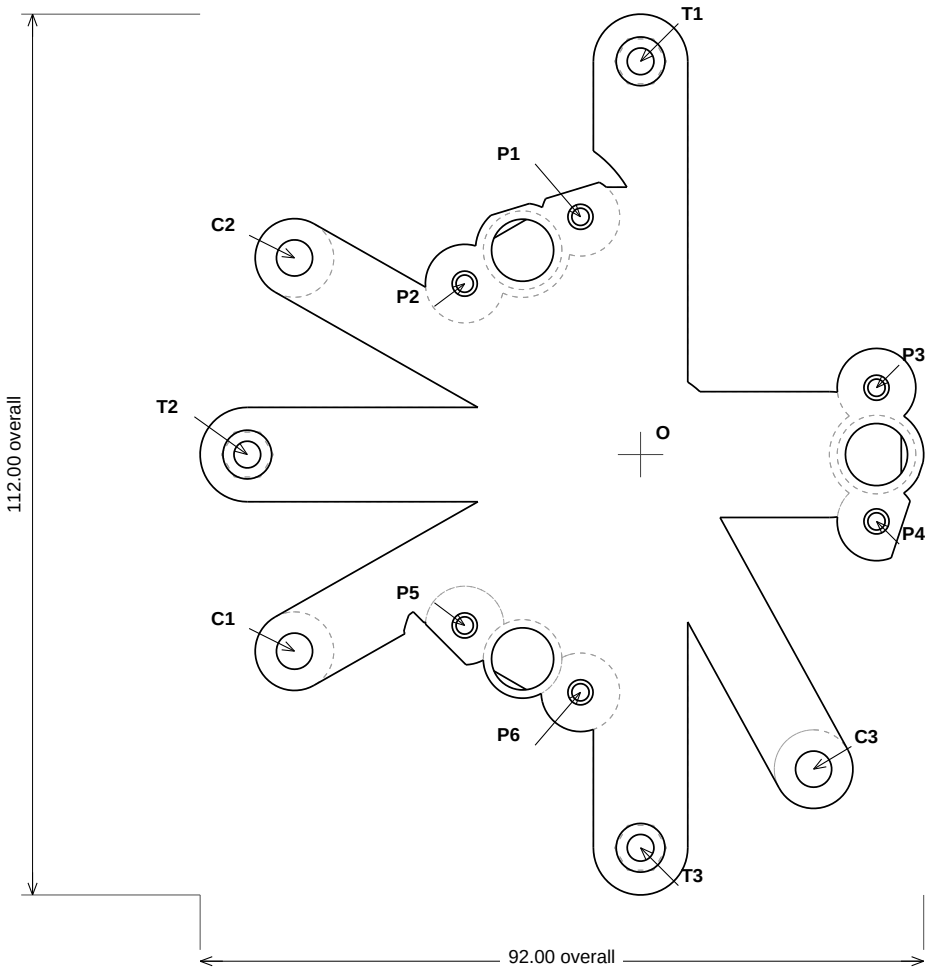
Fixed frame - front plate

All dimensions in mm | Fixed frame only | Right-hand views; use matching left-hand CAD

Print blank - rear face view along +Y

Solid lines: visible CAD edges. Dashed: opposite-side details.



Datum O = shaft axis: source X = -56.292, Z = 0. X is right; Z is up.

PETG plate: 6 base thickness

Tube pockets: 3 x Ø6.2 x 1.4 deep, rear.
Captured nut pockets: AF5.7 x 2.5 deep, front.
Minimum axial web between pockets: 2.1.

Hole centres from datum O

ID	X	Z	Feature
T1	0	50	Tie Ø3.4
T2	-50	0	Tie Ø3.4
T3	0	-50	Tie Ø3.4
C1	-44	-25	Chassis Ø4.6
C2	-44	25	Chassis Ø4.6
C3	22	-40	Chassis Ø4.6

M2 insert pilots - actual print blank

ID	X	Z
P1	-7.639	30.231
P2	-22.361	21.731
P3	30	8.5
P4	30	-8.5
P5	-22.361	-21.731
P6	-7.639	-30.231

6 x Ø3.2 insert pilots, 4 deep; Ø2.2 through.
Check insert fit on a print coupon before the batch.
Existing cup and cable-anchor interfaces are retained.

Print part: Fixed_Star_Plate_Print_Blank (+ _Left).

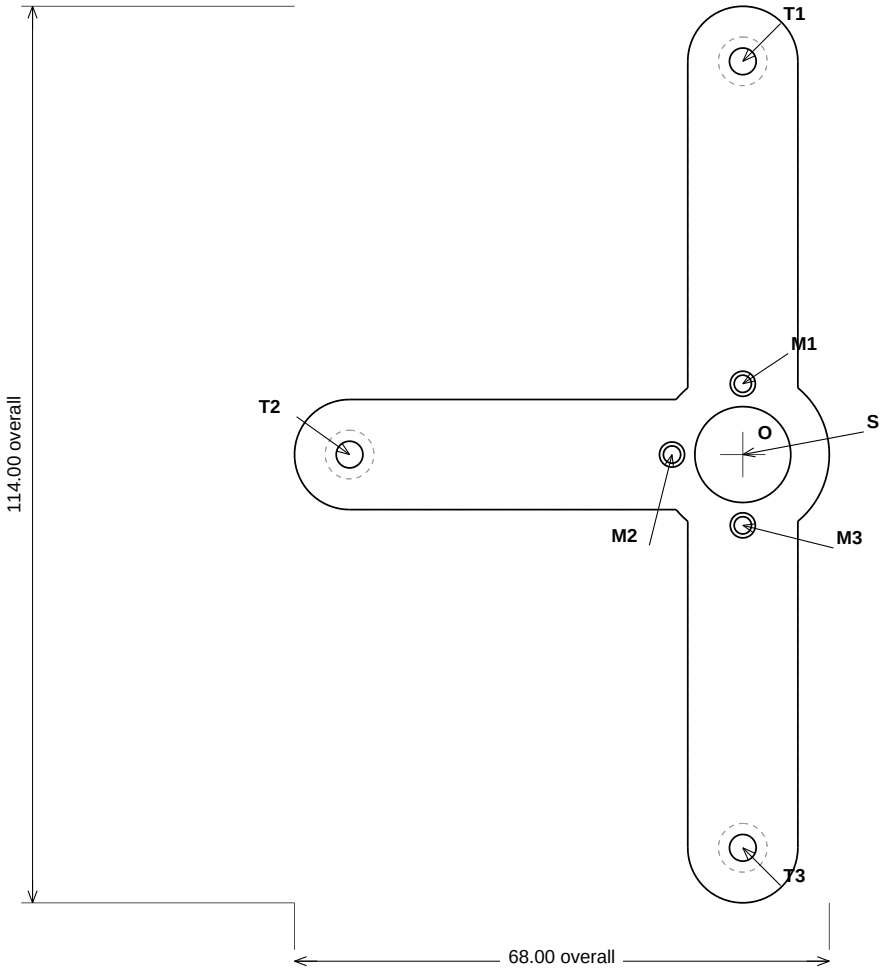
Rover quantity: 3 right-hand + 3 left-hand.

Fixed frame - rear support plate

All dimensions in mm | Fixed frame only | Right-hand views; use matching left-hand CAD

Print blank - rear face view along +Y

Use the supplied contour. Do not scale this sheet to cut the outline.



Datum O and spacer-hole centres are shared with the front plate.

PETG plate: 8 thickness

Tube pockets: 3 x Ø6.2 x 1.6 deep, front.
Shaft clearance: Ø12.2. Tie holes: Ø3.4.
M2 insert pilots: 3 x Ø3.2 x 4 deep; Ø2.2 through.

Hole centres from datum O

ID	X	Z	Feature
T1	0	50	Tie Ø3.4
T2	-50	0	Tie Ø3.4
T3	0	-50	Tie Ø3.4
S	0	0	Shaft Ø12.2
M1	0	9	M2 pilot
M2	-9	0	M2 pilot
M3	0	-9	M2 pilot

Print the print-blank part, not the installed-insert envelope.
Installed M2 insert seats are Ø3.5. The print blank leaves Ø3.2 pilots for fit trials.
Print the Y = -157 face on the bed for the right-hand plate.
Tube pockets face up.
Keep the original M2 collar hardware and shaft datums.

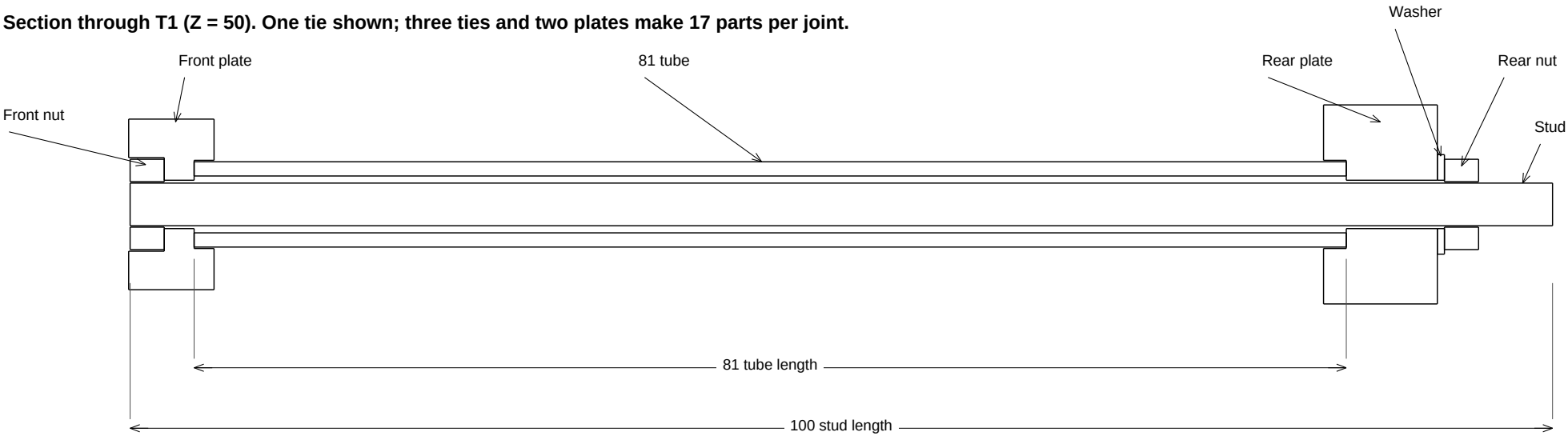
Print part: Rear_Support_Plate_Print_Blank (+ _Left).

Rover quantity: 3 right-hand + 3 left-hand.

Frame stack and cut list

All dimensions in mm | Fixed frame only | Right-hand views; use matching left-hand CAD

Section through T1 (Z = 50). One tie shown; three ties and two plates make 17 parts per joint.



Tube seats: 1.4 front + 1.6 rear. Clear face gap: 78. Minimum front pocket web: 2.1.

M3 x 0.5 purchased all-thread and nuts. CAD thread envelopes do not define thread manufacture.

Part / material	Per joint	Rover	Cut / fit
Front print blank	1	6	6 base; 3 RH + 3 LH
Rear print blank	1	6	8 thick; 3 RH + 3 LH
Aluminium tube Ø6 / Ø4	3	18	81 finished; deburr both ends
M3 x 0.5 all-thread stud	3	18	100 finished; deburr thread ends
M3 hex nut, AF5.5	6	36	3 captured front + 3 rear / joint
M3 washer	3	18	Rear side; use CAD washer size

Assembly

1. Fit inserts after a coupon check.
2. Seat front nuts; set stud ends flush.
3. Fit tubes, suspension and shaft.
4. Add rear plate, washers and nuts.
5. Tighten evenly; check free travel.
6. Use the existing three M4 chassis bolts.

Tube stock for these six frames: 18 x 81 = 1458 finished; allow 36 total saw kerf -> 1494 stock.

Total project M3 rod allocation: 3000 (2744 finished + 48 kerf; existing rod use included).

Prototype verification pending: print fits, 30 kg total-rover load, creep, fastener retention and loaded travel. No FEA or load rating.